Homework 2: Solutions

More Time, More Money

1 Financial Literacy

1.1 Definitions

- **Equity:** Residual claim on an asset's cash flows after all debt is paid.
- **Shares:** A security (a tradable financial claim on cash flows) representing proportional ownership of a corporation's equity associated with a claim on its assets and earnings.
- **Derivatives:** A security that can derive its payoff from the price of other assets, securities or underlying variables (such as a rate, an index, etc.)
- **Discount rate:** Rate of return used to convert future cash flows into present value, reflecting the time value of money and risk.
- **Arbitrage opportunity:** A trade that profits by exploiting price differences without taking any risk. In perfect markets, no arbitrage opportunities exist.
- **Expected value:** Anticipated value for a given investment. In probability, calculated by multiplying each possible outcome by its likelihood and summing over all outcomes.
- **Yield curve:** A plot of the interest rates (yields) at a set point in time of similar bonds but differing contract lengths (maturities).
- **Spread:**
  - **Bid-Ask Spread**: difference between the highest price a buyer is willing to pay and the lowest price a seller is willing to sell.
  - **Yield Spread**: difference in yields between two different fixed-income securities.

1.2 Thought Questions

Most investments are either debt or equity. Assume that there is an entrepreneur has an opportunity that requires an investment of X (that she must raise from investors) and has payout of Y in one year. If markets are competitive and the rate of interest is r, write a formula that gives the profit of the entrepreneur as a function of r, X and Y.

Profit is $\pi = Y - (1 + r)X$

Suppose that the entrepreneur decides to borrow, what contract will she sign?

She will sign a one-year debt contract with face value $F = (1 + r)X$

Suppose that the entrepreneur issues equity, what is her share of the firm.

Investors will buy a share α in the project to receive αY next year. As the market is competitive, the amount invested will be $X = \frac{\alpha Y}{1+r}$, so $\alpha = \frac{X(1+r)}{Y}$. The entrepreneur keeps the remaining fraction of $1 - \alpha = 1 - \frac{X(1+r)}{Y} = \frac{Y - X(1+r)}{Y}$.

Does it matter whether she issues debt or equity?

With debt the entrepreneur profit is $\pi = Y - F = Y - (1 + r)X$

With equity the entrepreneur profit is $\pi = (1 - \alpha)Y = \frac{Y - X(1+r)}{Y} \times Y = Y - X(1 + r)$.

The two are equal. Debt versus equity changes how the Y cash flow is split between the entrepreneur and investors. Under standard assumptions (e.g. competitive market), the total value stays the same.

2 Present Value

Depending on the assumptions you make (especially the indexing), your numerical results may differ from the solutions, but the underlying logic should remain the same.

2.1 Problem 2A — The 2013 Version

Early Tuesday October 1, 2013, Tesla Motors shares traded at \$190, thus its equity value was in excess of 20 billion dollars. This year it expects to sell 20,000 cars at \$70,000 each.

(A) Supposing a generous 25% net profit per car, how many cars per year does Tesla have to sell each year from year 5 to 20 years hence (assume linear growth in car sales from year 1 to 5) to justify a market cap of 20 billion dollars? Assume that the interest rate is 5%. (*Hint: compute net profits each year and then discount.*)

Assume January 1, 2013, is $t = 0$. At the end of 2013, so when $t = 1$, Tesla sells $20,000$ cars and receives the first cash flow. The growth period is $t \in [1, 5]$, corresponding to 2013–2017. Then, from 2018 to 2032, so for $t \in [6, 20]$, the number of sales is constant.

Profit per car is $0.25 \times 70,000 = 17,500$. Number of cars sold each year is N_t, with $N_1 = 20,000$. For the first five years with $t \in [1, 5]$, under linear growth, we get:

$$N_t = 20,000 + \frac{t-1}{4}(N_5 - 20,000)$$

For $t \in [6, 20]$, we have $N_t = N_5$. With the 20 billion market cap, N_5 should solve:

$$17,500 \times \left(\sum_{t=1}^{5} \frac{N_t}{(1.05)^t} + \sum_{t=6}^{20} \frac{N_5}{(1.05)^t} \right) = 20 \times 10^9$$

Solving this we get $N_5 \approx 107,678$ cars per year. The annual increase in cars sold each year should be approximately $22,000$, which implies a growth rate around 52%.

(B) How likely is it that there is enough demand for \$70,000 electric cars that Tesla will reach that target number of sales?

This is highly unlikely. To see why, we can look at sales data for the Model S in the US. This comes from a later period (2015–2023), but it is informative for intuition. In Figure 1, we see that sales are stagnant over time. The large spike in 2022 reflects the relaunch of the Model S rather than a sustained increase in demand. This already suggests, by the way, that a linear growth model is a poor description of actual sales

dynamics. But even if we redistribute the excess sales from 2022 across the whole window, the implied average annual increase remains far below the 22,000 additional cars per year required to reach the target sales level.[1]

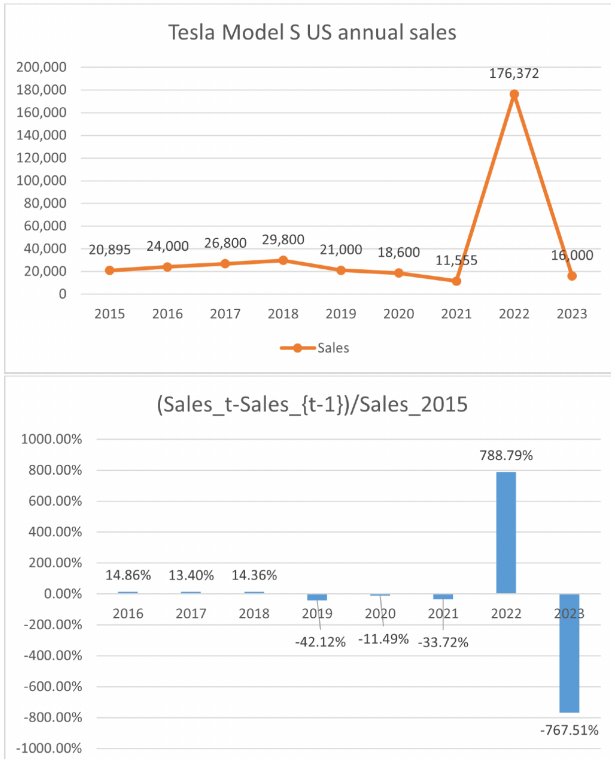

Figure 1: Tesla Model S Sales.

(C) What data could you use to give a more accurate answer to the question (you do not need to find the data)?

Besides actual sales data, some forecasts or market analyses by experts may also be helpful, other sources may include consumer surveys, sales of other brands, etc.

(D) Suppose five years hence Tesla has 30% of the luxury car sales market (which is 200,000 units of its current model per year). Tesla has announced that it intends to introduce a \$30,000 sedan in the next three years. How many such cars at the same net profit margin would it need to produce to justify the market cap it has reached recently (from year 3 to 20)?

Assume Tesla introduces the new sedan on January 1, 2016. It means that there is no cash flow associated with it in 2013–2015, so for $t \in [1, 3]$. For $t \in [4, 20]$, we assume

[1] More information on annual sales and growth rates can be found at Tesla Data.

Tesla sells a fixed number S of sedans per year; an alternative would be to solve for the total number of sedans sold over that period. The associated present value is:

$$PV_S = 30,000 \times 0.25 \times S \left(\sum_{t=4}^{20} \frac{1}{(1.05)^t} \right)$$

For the luxury cars, we are told that after 5 years Tesla will sell $0.3 \times 200,000 = 60,000$ cars per year. Let's assume the same structure as before and get the present value:

$$PV_L = 17,500 \times \left[\sum_{t=1}^{5} \frac{1}{(1.05)^t} \Big(20,000 + 10,000(t-1) \Big) + \sum_{t=6}^{20} \frac{60,000}{(1.05)^t} \right]$$

We solve for S in $PV_S + PV_L = 20 \times 10^9$. We obtain $S \approx 116,424$ sedans per year.

(E) Suppose Tesla runs into development problems and instead of introducing the new model in 2016 it has to delay to 2018. How would that affect your calculations in (D)?

We need to change the indexing of the invention from $t = 4$ to $t = 6$. PV_S becomes:

$$PV_S = 30,000 \times 0.25 \times S \left(\sum_{t=6}^{20} \frac{1}{(1.05)^t} \right)$$

The rest is unchanged. Solving for S again, we now get that Tesla needs to sell approximately 139,418 sedans per year to justify the market cap.

(F) What do you think Tesla's stock price should be? What calculation about sales volumes and per car net income justifies your answer?

You are free to choose any scenario that makes sense to you. Here is an example focusing on luxury cars only. From Part B of this question, we can see that the "normal" growth rate is about 15% for 2015-2018. Assume that this is the rate we want. Let's adopt the same structure that we worked with, namely a start at $20,000$ cars sold in 2013, an increase for first 5 years and stagnation for the remaining years.

$$\begin{aligned} \text{Market Cap} \;=\; 17,500 \times & \left(\sum_{t=1}^{5} \frac{20,000 \times 1.15^{(t-1)}}{(1.05)^t} + \sum_{t=6}^{20} \frac{20,000 \times 1.15^4}{(1.05)^t} \right) \\ & \approx 6.99 \times 10^9 \end{aligned}$$

From the information about market value and price, we know there are around:

$$\frac{20 \times 10^9}{190} \approx 105,263,158 \text{ shares.}$$

Under this scenario the price of a stock should be around \$66.5:

$$P_T = (6.99 \times 10^9)/105,263,158 \approx 66.5$$

2.2 Problem 2B — The 2023 Version

On April 6, 2022, Tesla traded at \$350 a share; on January 6, 2023, it traded at \$114 a share. It has 3 billion shares outstanding.

(A) Supposing that Tesla makes a \$10,000 profit on each car it sells, and interest rates at 2.5% provide an estimate of the number of cars Tesla should sell over the next 10 years to justify the \$350 a share (*hint: pick a final sales number and assume a constant growth rate over the decade*).

Assume that January 1, 2022 is $t = 0$. Then, $t = 1$ is associated with the cash flow received at the end of 2022. The last cash flow at $t = 10$ is the one at the end of 2031. One way to solve the problem is to pick the number of sales for 2022 and then solve for g. Tesla delivered approximately 1.31 million cars in 2022.[2] The setup then becomes:

$$10,000 \times \left(\sum_{t=1}^{10} \frac{1.31 \times 10^6 \times (1+g)^{t-1}}{(1.025)^t} \right) = 350 \times 3 \times 10^9$$

Solving for g we get a growth rate of 46.9% per year. Tesla should sell $N = \sum_{t=1}^{10} 1.31 \times 10^6 \times (1 + 0.469)^{t-1} \approx 128$ million cars over the decade to justify the 350 share.

(B) Interest rates are now 4.5%; does that justify the decline in stock price?

We slightly modify the equation to get:

$$10,000 \times \left(\sum_{t=1}^{10} \frac{1.31 \times 10^6 \times (1.469)^{t-1}}{(1.045)^t} \right) = P_T \times 3 \times 10^9$$

Solving for P_T we get an implied price of approximately \$300, which doesn't justify the decline in stock price.

[2] You can find the data here.

(C) Keeping interests at 4.5%, how many fewer cars is Tesla now expected to sell if the stock price is \$114 a share (keeping the per-car profit constant)?

We want to solve for a new g:

$$10,000 \times \left(\sum_{t=1}^{10} \frac{1.31 \times 10^6 \times (1+g')^{t-1}}{(1.045)^t} \right) = 114 \times 3 \times 10^9$$

We obtain $g' \approx 0.265$. The corresponding number of cars is $N' = \sum_{t=1}^{10} 1.31 \times 10^6 \times (1+0.265)^{t-1} \approx 46.9$ million cars over the decade. This is $128 - 46.9 = 81.1$ million fewer cars expected to be sold compared to Part A.

(D) Keeping forecast sales of Part (A) and interest rates at 4.5%, by how much does the profit rate per car have to change to justify the \$114 sale price?

Solve the following equation:

$$\pi \times \left(\sum_{t=1}^{10} \frac{1.31 \times 10^6 \times (1.469)^{t-1}}{(1.045)^t} \right) = 114 \times 3 \times 10^9$$

This implies $\pi \approx 3,800$ dollars per car.

2.3 Problem 2C — The 2024 Version

Tesla is now trading in the \$240 per share range. This is a 50% increase since last year.

1. What macro-economic news might explain this very positive turn of events?

   On the macro level, investors' confidence may be boosted with the US real GDP growth stronger than expected in 2023 and manufacturing investment reaching historic highs. Meanwhile, investors may expect the FED to stop raising interest rates after seeing job growth cool down, inflation slow down, and an overall cool-down of the economy.

2. What Tesla-specific news might explain the same very positive turn of events?

   On the firm level, positive news includes the price cut of Model Y and Model 3 and the release of Cybertruck.

2.4 Problem 2D — The 2025 Version

Tesla is now trading at about \$380 per share range. This is an 80% increase since last year despite the fact that its 2024 sales were down 1.1% relative to 2023.

1. What macro-economic news might explain this very positive turn of events?

   In November 2024, consumer spending rose by 0.4 %. Additionally, the US economy experienced strong growth of 2.8% in the last quarter of 2024.

2. What Tesla-specific news might explain the same very positive turn of events?

   Tesla received several pieces of positive news: Donald Trump's election victory (due to the ties with Elon Musk), the completion of Tesla's Full Self-Driving technology, the announcement that older Tesla models will receive upgraded cameras, a reduction in charging costs at Tesla Superchargers, and the decision to open Tesla's charging network to vehicles from other manufacturers. Tesla sales in China have increased

3. Supposing Tesla starts selling a car at \$25,000 with a margin of 40%, how many of these cars would it have to sell to justify the increased market capitalization from \$600 billion to \$1.1 trillion (assuming sales of the other models remain constant)? You can discount the future at 0.95 per year.

   The market valuation at 600 billion was calculated with the previous Tesla models. Assuming their expected sales don't change, it means that the 500 billion increase in valuation is uniquely due to the new model. From the discount rate, we can derive an interest rate solving $d = 0.95 = \frac{1}{1+r} \implies r \approx 0.05$. Assuming that the increase in car sales should be realized in the next 10 years (but no growth rate g), we can get:

   $$25,000 \times 0.4 \times N \times \left(\sum_{t=1}^{10} \frac{1}{(1.05)^t} \right) = 500 \times 10^9$$

   Solving for N we get that Tesla should sell 6.5 million cars per year of the new model.

2.5 Problem 2E — The 2026 Version

Tesla is now trading at about \$450 per share range. This is a 20% increase from a year ago. Given that its automobile sales have been flat for the last two years, do you think it is it still a car company? Whatever you may think, your boss has asked you to provide a sales forecast for revenue from selling car at \$30,000 a car with a margin of 40% over the next 20 years that is consistent with a firm value of 1.5 trillion. Assume the cost of capital is 6%.

The annual sales forecast is:

$$12,000 \times N \times \left(\sum_{t=1}^{20} \frac{1}{(1.06)^t} \right) = 1.5 \times 10^{12}$$

Solving for N, we obtain roughly 10.9 million cars per year for a total of 218 million of cars over 20 years.

3 Debt vs. Equity

Two Caltech undergraduates (J&J) create a new windmill that is adaptable to extremely cold and wet climates. To develop this invention, they need 75 million dollars in capital for plant, equipment, and advances to power companies in Greenland and Nova Scotia. Net of variable costs (labor, supplies, shipping, etc.), the firm would clear 7 million dollars a year starting two years after starting for 20 years.

(A) What interest rate makes investors indifferent between starting the firm or not? Show that this is the internal rate of return of the project.

There is no return for $t \in \{0, 1\}$ and a 7 million return for $t \in \{2, 21\}$

Making the investors indifferent means that $NPV = 0$ and IRR is precisely defined as the rate that makes $NPV = 0$. To get it, solve:

$$\sum_{t=2}^{21} \frac{7}{(1+r)^t} = 75 \implies r^* \approx 6.11\%$$

(B) If the interest rate is 5% and there is no equity premium, show that J&J are indifferent between financing their capital needs by debt or by equity. What fraction of the firm's income value do the two inventors retain after the plant gets going?

Indifference is similar to 1.2. Debt financing implies issuing a loan with face value F that is paid off the value of the firm after two years, solving:

$$75 = \frac{F}{(1.05)^2} \implies F = 75(1.05)^2$$

Fiancing with equity implies that investors will owe α of the present value at the end

of year 2. At the start of the project it is worth the same than F:

$$75 = \frac{\alpha V_2}{(1.05)^2} \implies \alpha V_2 = 75(1.05)^2$$

We can use the info we have to actually calculate V_2:

$$V_2 = \sum_{k=0}^{19} \frac{7}{(1.05)^k} \approx 91.6 \text{ million}$$

Financing starts two years before. Using the above expression, α will solve:

$$75 = \frac{\alpha 91.6}{(1.05)^2} \implies \alpha = 0.903$$

The fraction retained by J&J is $1 - \alpha = 9.7\%$.

(C) Now suppose there is a 20% possibility that the government of Nova Scotia decides at the end of year 1 to ban the new windmill, reducing income by 50%. Show again that if markets are efficient, combinations of debt and equity do not affect the value of the firm.

At $t = 2$ there are two possible values associated with good (G) or bad (B) states:

$$V_2^G = \sum_{k=0}^{19} \frac{7}{(1.05)^k} \quad , \quad V_2^B = \sum_{k=0}^{19} \frac{3.5}{(1.05)^k}$$

At $t = 0$ the debt value is:

$$D_0 = \frac{0.8 \min(F, V_2^G) + 0.2 \min(F, V_2^B)}{(1.05)^2}$$

At $t = 0$ the equity value is:

$$E_0 = \frac{0.8[V_2^G - \min(F, V_2^G)] + 0.2[V_2^B - \min(F, V_2^B)]}{(1.05)^2}$$

With efficient market, the total value of the project at $t = 0$ is:

$$V_0 = D_0 + E_0 = \frac{0.8V_2^G + 0.2V_2^B}{(1.05)^2}$$

It doesn't depend on the level of F.

(D) What financial structure implies that after an adverse regulatory decision by the government of Nova Scotia, the company would belong to bondholders?

D is the amount borrowed. The project would belong to bondholders if:

$$V_2^B = \sum_{k=0}^{19} \frac{3.5}{(1.05)^k} \leq F = (1.05)^2 D$$

We get the threshold of $D^* = 41.6$ million. It means that if they borrow more than this amount, the project will belong to bondholders in the bad state.

(E) Suppose now that interest rates are 6%. What financial structure implies that after an adverse regulatory decision by the government of Nova Scotia, the company would belong to bondholders?

Same as before but the numerator is $(1.06)^k$. If yields a threshold of $D^* = 37.8$ millions.

(F) Suppose that J&J are risk averse. What financial structure are they likely to choose?

They will want to minimize the amount of debt to reduce the risk of being wiped out in bad states. They will therefore choose an amount D below the threshold D^*.